Tensor Math

list = [1,5,2,3]

list of lists = [[1,2,3,2],[1,2,3,4]]

list of list of lists = [[[[1,2,3],[1,2,3]],[[1,2,3],[1,2,3]]]

Matrix = [[4,2],
[5,1],
[8,2]] $M_{3\times2}$ its 2D

[[[1,5,6],
[3,2,1]],
[[...],
[...]],
[[...],
[...]]] → 3D

Tensors are Mathematically not Matrices but can be represented by them

~~Instead of ... th~~

n

$$\vec{a} \cdot \vec{b} = \sum_{i=1} a_i b_i$$

$$\begin{bmatrix} \text{Input} \end{bmatrix} \begin{bmatrix} W \\ e \\ i \\ g \\ h \\ t \end{bmatrix} +$$

neuron → vector

a layer is a Matrix of these vectors

$$\begin{bmatrix} \text{Input} \end{bmatrix} \begin{bmatrix} W & W & W & W \\ e & e & e & e \\ i & i & i & i \\ g & g & g & g \\ h & h & h & h \\ t & t & t & t \end{bmatrix} + \begin{bmatrix} b \\ i \\ a \\ s \end{bmatrix} = \begin{bmatrix} O \\ u \\ t \\ p \\ u \\ t \end{bmatrix}$$